AOA-KV: Attention Output-Aware KV Selection for Efficient Long-Context LLMs

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Abstract

Extensive computation and memory I/O incurred by attention have become a major bottleneck in large language models inference, especially in long-context scenarios. To mitigate this cost, prior work has explored dynamic KV selection schemes; however, they rely solely on query–key similarity and neglect the contribution of values to the attention output. In this work, we propose a novel selection strategy, Attention Output-Aware KV selection (AOA-KV), which jointly considers queries, keys, and values to minimize the output error under dynamic KV sparsity. Experimental results show that AOA-KV improves accuracy by up to 4.7% points on RULER and LongBench against previous art, while achieving a 1.8× end-to-end speedup over full KV access at a 32K context.

1 Introduction

With the growing interest in long-context tasks such as multi-document question answering and repository-level code understanding (Bai et al., 2023; Jimenez et al., 2023), the context length of large language models (LLMs) has expanded rapidly (Achiam et al., 2023; Reid et al., 2024; Yang et al., 2025a). However, due to the design of the attention mechanism, each token generation step requires accessing all preceding key–value (KV) pairs, leading to substantial memory traffic and making KV-cache access a primary bottleneck for inference latency in long-context scenarios.

To address this bottleneck, recent work has explored token pruning and selection methods for KV caches. Among them, QUEST (Tang et al., 2024) groups a set of key–value pairs into pages and selectively retrieves KV pages based on query–key similarity, achieving over a 7× self-attention speedup with minimal accuracy loss. Its effectiveness has made QUEST a core building block for many subsequent studies (Liu et al., 2025; Yang et al., 2025b; Behnam et al., 2025).

Despite its widespread adoption, query–key similarity provides only a partial view of the attention. Since the attention output is computed as a weighted sum over values, selection strategies that rely solely on query–key matching implicitly ignore value-level contributions. As a result, methods such as QUEST may select KV pages that are suboptimal in approximating the final output. In this work, we formally show that minimizing attention output error requires a selection criterion that goes beyond query–key similarity alone.

Motivated by this insight, we propose AOA-KV (Attention Output-Aware KV selection), a page-level selection strategy that explicitly minimizes attention output error by jointly accounting for queries, keys, and values. AOA-KV introduces only negligible overhead compared to QUEST, yet substantially improves selection quality under the same KV cache budget. Experiments across multiple LLMs on long-context benchmarks such as RULER and LongBench demonstrate that AOA-KV consistently outperforms prior query–key–only baseline, QUEST, achieving accuracy gains of up to 4.7% points. Moreover, by leveraging precomputed value statistics, AOA-KV preserves QUEST's efficiency, achieving up to a 1.8× end-to-end speedup over full KV attention at a 32K context length. Beyond introducing a new state-of-the-art KV selection method, our work provides a principled interpretation of KV selection that may serve as a foundation for future research.

2 Related Work

To reduce inference costs in long-context tasks, numerous token pruning and selection methods have been proposed. Static token pruning approaches (Zhang et al., 2023; Xiao et al., 2023; Li et al., 2024; Guo et al., 2024) evicts tokens based on token-level importance scores, but inevitably incur irreversible information loss and struggle to adapt to the chang-

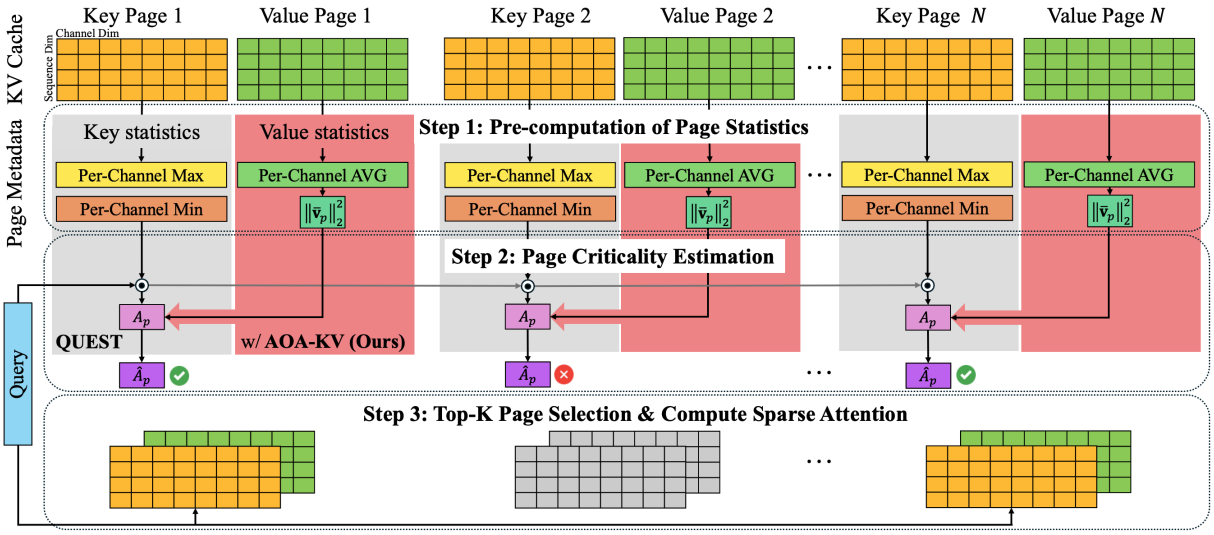

Figure 1: **Overall pipeline of AOA-KV. Step 1:** When a KV page is full, channel-wise min/max keys and the squared L_2-norm of the average value are pre-computed. **Step 2:** During decoding, page criticality is estimated not only from the current query and key statistics but also from value statistics. **Step 3:** Only the Top-K pages with the highest criticality are retrieved to perform sparse attention.

ing relevance of tokens during generation.

More recently, dynamic KV selection methods (Tang et al., 2024; Liu et al., 2025; Chen et al., 2024; Sun et al., 2024) have emerged as an alternative. These methods retain the full KV cache in GPU memory and dynamically retrieve query-relevant KV pages during generation, enabling acceleration without discarding information. Although this design does not reduce memory usage, it offers a favorable trade-off by improving inference efficiency while largely preserving accuracy.

AOA-KV also builds upon the page-level dynamic selection, which retains hardware efficiency as well. AOA-KV introduces a more principled, output-aware filtering strategy that further improves KV selection quality without incurring additional information loss.

3 Motivation

3.1 Limitation of Query-Key based Key-Value Selection

QUEST (Tang et al., 2024) is a representative baseline for dynamic KV selection. Its central idea is to efficiently approximate the similarity between a query and all keys within a page using a compact key representation, thereby avoiding exact attention score computation over the full KV cache.

Once accumulated keys and values fill a page p, QUEST summarizes the keys by computing channel-wise minimum and maximum values over $\mathbf{K}_p$, the keys in p. This produces two representative vectors, $\mathbf{k}_{\min,p}$ and $\mathbf{k}_{\max,p}$, which are stored for future queries (Figure 1, Step 1):

$$\begin{aligned} \mathbf{k}_{\min,p} &\triangleq \mathrm{MinPool}(\mathbf{K}_p) \in \mathbb{R}^{d_h}, \\ \mathbf{k}_{\max,p} &\triangleq \mathrm{MaxPool}(\mathbf{K}_p) \in \mathbb{R}^{d_h}, \end{aligned} \tag{1}$$

where d_h denotes the head dimension. For a given query $\mathbf{q}_t$ at timestep t, the criticality A_p of page p is estimated as:

$$A_p \triangleq \max\left(\mathbf{q}_t \mathbf{k}_{\min,p}^\top,\ \mathbf{q}_t \mathbf{k}_{\max,p}^\top\right). \tag{2}$$

Under a fixed computation budget, QUEST selects the top-k pages with the highest estimated criticality for the subsequent attention.

While this strategy is effective at identifying pages whose keys are highly aligned with the query, it optimizes a proxy objective based solely on query-key similarity. However, the attention mechanism ultimately produces outputs through a weighted aggregation over values, raising a fundamental question: *does selecting pages whose keys best match the query necessarily minimize the discrepancy in the final attention output?*

3.2 Importance of Output-aware Key-Value Selection

Motivated by this observation, we formalize KV selection as an output-preserving problem, which **selects a subset of KV pages whose attention output closely matches the original output of full KV cache attention.**

Let $\mathbf{o}_{\text{full}} \in \mathbb{R}^{d_h}$ denote the attention output computed using a full KV cache, and $\mathbf{o}_{\text{sel}} \in \mathbb{R}^{d_h}$ the output obtained using a selected subset of KV pages. Under this formulation, KV selection can be defined as the problem of finding an optimal page set $\mathcal{S}_k$ that minimizes the squared L_2 error between the two outputs:

$$E = \|\mathbf{o}_{\text{full}} - \mathbf{o}_{\text{sel}}\|_2^2 . \tag{3}$$

Assuming that the selected pages capture the dominant attention mass, this objective can be further approximated as follows (please refer to Appendix A for more details):

$$\min_{\mathcal{S}_k} E \approx \max_{\mathcal{S}_k} \sum_{p \in \mathcal{S}_k} \exp\left(\frac{A_p}{\sqrt{d_h}}\right)^2 \|\bar{\mathbf{v}}_p\|_2^2 . \tag{4}$$

Here, $\bar{\mathbf{v}}_p$ denotes the per-channel average of the value vectors $\mathbf{V}_p$ within page p. Crucially, this formulation shows that from an output error minimization perspective, *page-level value information is as important as query-key based estimation*, and must be explicitly considered during KV selection.

4 Proposed Idea: AOA-KV

To realize the objective in Eq. 4 with minimal additional overhead, we propose *Attention Output-Aware KV selection (AOA-KV)*. AOA-KV follows the same page-based KV selection as QUEST, thereby preserving hardware efficiency and ease of acceleration, while being explicitly designed to improve downstream task accuracy from an output-preserving perspective.

Figure 1 highlights the additional components in red. The core idea is to select KV pages by explicitly accounting for the contribution of value vectors within each page, by relying on *value-aware criticality,* $\hat{A}_p$ for each page p:

$$\hat{A}_p \triangleq \exp\left(\frac{A_p}{\sqrt{d_h}}\right)^2 \|\bar{\mathbf{v}}_p\|_2^2 . \tag{5}$$

Compared to A_p, the key distinction of $\hat{A}_p$ is the incorporation of the *average value magnitude* within each page as a weighting factor. Importantly, the term $\|\bar{\mathbf{v}}_p\|_2^2$ is query-independent and can be computed once when a page is finalized, then reused across all subsequent decoding steps (Figure 1, Step 2).

As a result, AOA-KV introduces only minimal overhead compared to QUEST. Page selection requires a single per-page scalar multiplication by $\|\bar{\mathbf{v}}_p\|_2^2$, along with lightweight element-wise operations such as exponentiation and normalization by $\sqrt{d_h}$. These costs are negligible relative to full attention computation, making AOA-KV both efficient and practically deployable.

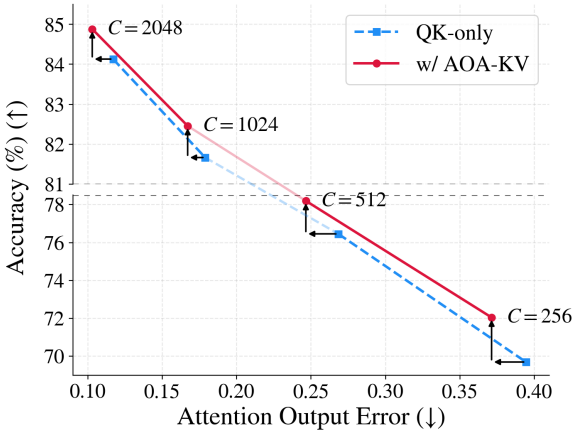

Figure 2: **Correlation between Attention Output Error (AOE) and Accuracy.** We evaluate Llama-3.1-8B-Instruct on RULER-32K across various KV budgets (C). **AOA-KV** (orange) consistently achieves lower AOE than the **QK-only** baseline (blue), which leads to superior long-context accuracy.

5 Experiments

Experimental Setup. We evaluate six LLMs: the MHA-based *LongChat-v1.5-7B* (Li et al., 2023), and the GQA-based *Mistral-7B-v0.2-Instruct* (Jiang et al., 2023), *Llama-3.1-8B/3.2-3B-Instruct* (Dubey et al., 2024), and *Qwen-2.5-7B/3B-Instruct* (Yang et al., 2024). We compare against **Full KV Cache** attention and **QUEST** (Tang et al., 2024). Experiments are conducted across a wide range of sequence lengths and KV budgets (C) on RULER (Hsieh et al., 2024) and LongBench (Bai et al., 2023) benchmarks. The details are available in Appendix B.

5.1 Importance of Output Error Reduction

To quantitatively validate the theoretical formulation derived in Section 3.2, we measure the attention output error (AOE) and analyze how changes in AOE correspond to accuracy. On RULER-32K, we compare the **QK-only** baseline (QUEST) and **AOA-KV** under various KV budgets (C) using Llama-3.1-8B-Instruct. As shown in Figure 2, AOA-KV consistently achieves lower AOE, which closely corresponds to improvements in accuracy. Specifically, at $C = 256$, AOA-KV reduces AOE by approximately 5.9% while simultaneously im-

Table 1: **Comprehensive accuracy evaluation on the RULER and LongBench benchmarks.**

Model	Method	RULER 8K (4K) 256	512	1024	2048	16K (8K) 256	512	1024	2048	32K (16K) 256	512	1024	2048	LongBench 256	512	1024
Llama 3.1-8B	**Full KV**	93.46				91.37				85.66				49.4		
	QUEST	81.84	88.30	91.64	92.75	**76.64**	80.67	85.70	88.20	69.70	76.46	81.67	84.13	45.7	47.3	**48.9**
	w/ AOA-KV	**83.77**	**88.58**	**91.90**	**93.21**	76.6	**82.92**	**86.7**	**89.41**	**72.05**	**78.20**	**82.46**	**84.88**	**46.2**	**48.3**	48.8
Llama 3.2-3B	**Full KV**	80.78				78.72				74.61				44.2		
	QUEST	69.79	75.25	78.46	80.92	57.94	**69.02**	73.35	76.56	49.51	59.07	67.21	70.03	40.0	42.5	44.3
	w/ AOA-KV	**71.11**	**76.88**	**80.51**	**81.95**	**59.44**	69.01	**74.37**	**77.84**	**52.24**	**61.37**	**67.32**	**71.65**	**40.2**	**43.2**	**44.6**
Qwen 2.5-7B	**Full KV**	91.54				92.39				90.26				48.7		
	QUEST	78.15	83.29	87.93	92.29	**73.32**	79.15	84.78	89.87	66.28	74.48	**79.67**	83.97	41.1	44.4	46.7
	w/ AOA-KV	**78.26**	**84.13**	**90.49**	**92.61**	73.27	**79.19**	**86.58**	**90.63**	**67.64**	**74.74**	78.82	**84.82**	**41.6**	**45.3**	**47.0**
Qwen 2.5-3B	**Full KV**	87.62				84.93				78.02				41.1		
	QUEST	65.85	73.30	76.87	84.46	59.45	66.97	72.86	77.31	44.46	58.24	66.04	71.15	**30.9**	**36.3**	**39.1**
	w/ AOA-KV	**68.42**	**74.37**	**78.48**	**85.27**	**61.69**	**70.36**	**75.43**	**79.32**	**49.14**	**61.67**	**67.71**	**71.76**	**30.9**	35.8	**39.1**
Long Chat-7B	**Full KV**	85.66				80.82				70.25				34.6		
	QUEST	80.5	**84.73**	85.16	85.44	68.67	75.00	79.52	**80.37**	57.38	**64.30**	**67.88**	70.68	33.2	34.5	**35.3**
	w/ AOA-KV	**80.82**	84.52	**85.41**	**85.56**	**68.98**	**75.69**	**80.23**	80.18	**58.60**	63.36	67.29	**70.81**	**33.5**	**34.9**	**35.3**
Mistral 7B	**Full KV**	94.17				89.42				86.78				42.7		
	QUEST	84.38	**91.51**	92.77	93.97	76.42	82.26	86.85	88.05	66.70	74.75	78.89	84.83	37.1	40.4	42.1
	w/ AOA-KV	**87.70**	91.45	**93.13**	**94.26**	**78.36**	**85.13**	**87.24**	**88.46**	**69.69**	**75.99**	**81.52**	**84.90**	**37.6**	**41.1**	**42.5**

proving accuracy by about 3.4% points. These results empirically corroborate that minimizing AOE via the proposed objective (Eq. (4)) is effective for preserving long-context accuracy.

5.2 Accuracy Evaluation

Table 1 shows that AOA-KV achieves overall accuracy improvements over the QK-only baseline across multiple models under various sequence lengths and KV budget configurations. Specifically, on RULER, for 4 out of 6 models including Llama-3.1-8B, accuracy gains are observed in at least 11 out of 12 configurations. This trend even holds on LongBench, where for 3 out of 6 models, AOA-KV achieves accuracy improvements across all 3 configurations. These results indicate that leveraging value statistics effectively complements existing query-key-based selection metrics.

5.3 Efficiency Evaluation

Figure 3 shows that AOA-KV achieves substantial speedups over the Full KV cache with negligible overhead compared to the QK-only baseline. On a single NVIDIA L40S GPU, we measure end-to-end decoding latency for LongChat-7B($C = 1024$) using a FlashInfer (Ye et al., 2025)-based custom CUDA kernel, and find that AOA-KV exhibits nearly identical latency to QUEST.

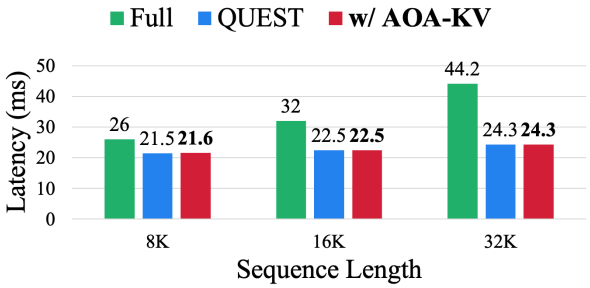

Figure 3: **End-to-end latency evaluation on LongChat-7B-v1.5-32K** ($C = 1024$).

This is enabled by pre-computed value statistics, which minimize additional computation. As a result, AOA-KV achieves the end-to-end inference speedups of 1.4× and 1.8× over the Full KV baseline at 16K and 32K contexts, respectively.

6 Conclusion

We propose **AOA-KV**, an output-aware KV selection that incorporates value contributions overlooked in prior dynamic KV retrieval to minimize attention output error. This approach achieves accuracy gains of up to 4.7% points and a 1.8× speedup with negligible overhead. By establishing a **principled output-aware perspective** on page-level selection, our work provides a key foundation for future efficient long-context inference research.

Limitations

Integration with Orthogonal Efficiency Strategies. This work focuses on deriving a principled metric for selection in page-level KV retrieval; consequently, we did not perform extensive validation on subsequent variants that incorporate additional computation reduction strategies into QUEST (Tang et al., 2024). Recently, various variants (Liu et al., 2025; Yang et al., 2025b; Behnam et al., 2025) have been proposed to enhance computation and retrieval efficiency by orthogonally combining strategies such as KV selection reuse across adjacent queries, channel pruning, and token eviction with QUEST's selection mechanism. While further validation is required regarding the performance and efficiency when combined with these strategies, these methods fundamentally follow QUEST's selection strategy at the core selection stage. Therefore, in this work, we focused our comparison on QUEST itself as the core selection mechanism; however, systematically analyzing the effects of our output-aware approach when combined with these additional efficiency techniques remains a promising direction for future research.

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A Derivation of KV Page Selection Metric for Attention Output Error Minimization

This section provides a detailed derivation of the objective function introduced in Section 3.2. Our goal is to identify the optimal subset of KV pages $\mathcal{S}_k$ that minimizes the attention output error under a limited budget.

A.1 Problem Definition

Consider the attention output error in a single attention head. Let $\mathbf{o}_{\text{full}} \in \mathbb{R}^{d_h}$ and $\mathbf{o}_{\text{sel}} \in \mathbb{R}^{d_h}$ denote the attention output vectors computed using the full KV cache and the selected page subset $\mathcal{S}_k$, respectively. We define the attention output error caused by KV page selection as the squared L_2 norm of the difference between these two vectors:

$$E \triangleq \|\mathbf{o}_{\text{full}} - \mathbf{o}_{\text{sel}}\|_2^2 \tag{6}$$

To identify the factors contributing to this error, we decompose Eq. (6) along the head dimension d_h. For each channel $l \in \{1, \ldots, d_h\}$, the error is expressed as the sum of the squared differences of the scalar components:

$$E = \sum_{l=1}^{d_h} \left(o_{\text{full}}^l - o_{\text{sel}}^l \right)^2 \tag{7}$$

A.2 Approximation of the Error Term

Let $\mathcal{U}$ denote the set of all pages, and let $\mathcal{S}_k \subset \mathcal{U}$ be the selected subset. We denote the set of token indices belonging to the selected pages as $\mathcal{I}(\mathcal{S}_k)$.

Expanding Eq. (7), E takes a form that involves the discrepancy in softmax normalization between the full and selected caches (for notational simplicity, we omit the scaling factor $\sqrt{d_h}$ in the subsequent derivation, specifically Eqs. (8)–(10)):

$$E = \sum_{l=1}^{d_h} \Bigg(\sum_{i \in \mathcal{I}(\mathcal{S}_k)} \left[\frac{\exp(\mathbf{q}\mathbf{k}_i^\top)}{\sum_{j \in \mathcal{U}} \exp(\mathbf{q}\mathbf{k}_j^\top)} - \frac{\exp(\mathbf{q}\mathbf{k}_i^\top)}{\sum_{j \in \mathcal{I}(\mathcal{S}_k)} \exp(\mathbf{q}\mathbf{k}_j^\top)} \right] v_i^l + \sum_{i \in \mathcal{I}(\mathcal{S}_k^c)} \left(\frac{\exp(\mathbf{q}\mathbf{k}_i^\top)}{\sum_{j \in \mathcal{U}} \exp(\mathbf{q}\mathbf{k}_j^\top)} \right) v_i^l \Bigg)^2 \tag{8}$$

To make this formulation tractable, we introduce an approximation assumption. Specifically, assuming that the selected pages capture the dominant attention mass, we consider the difference in softmax normalization terms to be negligible:

$$\frac{\exp(\mathbf{q}\mathbf{k}_i^\top)}{\sum_{j \in \mathcal{U}} \exp(\mathbf{q}\mathbf{k}_j^\top)} \approx \frac{\exp(\mathbf{q}\mathbf{k}_i^\top)}{\sum_{j \in \mathcal{I}(\mathcal{S}_k)} \exp(\mathbf{q}\mathbf{k}_j^\top)} \tag{9}$$

Under this assumption, the first term vanishes, and the approximation error simplifies to a term corresponding to the tokens $\mathcal{I}(\mathcal{S}_k^c)$ in the non-selected KV pages:

$$E \approx \sum_{l=1}^{d_h} \left(\sum_{i \in \mathcal{I}(\mathcal{S}_k^c)} \left(\frac{\exp(\mathbf{q}\mathbf{k}_i^\top)}{\sum_{j \in \mathcal{U}} \exp(\mathbf{q}\mathbf{k}_j^\top)} \right) v_i^l \right)^2 \tag{10}$$

Minimizing the error term caused by unselected pages in the above equation can be approximated by a surrogate objective that maximizes the attention output contribution of the selected pages $\mathcal{S}_k$, which simplifies as follows:

$$\begin{aligned} \min_{\mathcal{S}_k} E &\approx \max_{\mathcal{S}_k} \sum_{l=1}^{d_h} \left(o_{\text{sel}}^l \right)^2 \\ &= \max_{\mathcal{S}_k} \left\| \mathbf{o}_{\text{sel}} \right\|_2^2 \\ &= \max_{\mathcal{S}_k} \left\| \sum_{p \in \mathcal{S}_k} \mathbf{o}_p \right\|_2^2 \end{aligned} \tag{11}$$

A.3 Derivation of the Page-Level Metric

To derive a concrete page-level selection metric, we analyze the term $\left\| \sum_{p \in \mathcal{S}_k} \mathbf{o}_p \right\|_2^2$ from the perspective of page-level contributions. The output contribution of a single page p can be expressed as the sum of the token-level contributions belonging to that page:

$$\mathbf{o}_p = \sum_{i \in \mathcal{I}(p)} \alpha_i \mathbf{v}_i \tag{12}$$

where $\mathcal{I}(p)$ denotes the set of token indices in page p, and α_i and $\mathbf{v}_i$ represent the attention weight and value vector for token i, respectively.

Given that the KV cache is managed at the page level, we adopt the approach from QUEST (Tang et al., 2024) and introduce $A_p = \max(\mathbf{q}_t \mathbf{k}_{\text{min},p}^\top, \mathbf{q}_t \mathbf{k}_{\text{max},p}^\top)$ to approximate the attention weight for page p. Applying this approximation, the attention weight α_i assigned to token i in page p is estimated as:

$$\alpha_i = \frac{\exp(\mathbf{q}\mathbf{k}_i^\top/\sqrt{d_h})}{Z} \approx \frac{\exp(A_p/\sqrt{d_h})}{Z}, \quad \forall i \in \mathcal{I}(p) \tag{13}$$

Here, $Z = \sum_{j\in\mathcal{U}} \exp(\mathbf{q}\mathbf{k}_j^\top/\sqrt{d_h})$ is the normalization term for the softmax operation. For notational convenience, let us define $w_p \triangleq \exp(A_p/\sqrt{d_h})$. Using this, we derive the contribution vector $\mathbf{o}_p$ of page p to the final output as follows:

$$\begin{aligned} \mathbf{o}_p &= \sum_{i\in\mathcal{I}(p)} \alpha_i \mathbf{v}_i \\ &\approx \sum_{i\in\mathcal{I}(p)} \frac{w_p}{Z}\mathbf{v}_i \\ &= \frac{w_p}{Z}\left(\sum_{i\in\mathcal{I}(p)} \mathbf{v}_i\right) \\ &= \frac{P}{Z}\cdot w_p \cdot \bar{\mathbf{v}}_p \end{aligned} \tag{14}$$

where $P = |\mathcal{I}(p)|$ is the page size, and $\bar{\mathbf{v}}_p = \frac{1}{P}\sum_{i\in\mathcal{I}(p)} \mathbf{v}_i$ is the mean value vector of the page.

Substituting these approximations into the objective function in Eq. (11) yields:

$$\begin{aligned} \left\|\sum_{p\in\mathcal{S}_k} \mathbf{o}_p\right\|_2^2 &\approx \left\|\sum_{p\in\mathcal{S}_k} \frac{P}{Z}\cdot w_p \cdot \bar{\mathbf{v}}_p\right\|_2^2 \\ &= \left(\frac{P}{Z}\right)^2 \left\|\sum_{p\in\mathcal{S}_k} w_p \cdot \bar{\mathbf{v}}_p\right\|_2^2 . \end{aligned} \tag{15}$$

Since the scaling factor $(P/Z)^2$ is a positive constant common to all page sets, it does not affect the ranking for selection and can be omitted. Excluding the constant term and assuming that the contributions between pages are independent (thereby eliminating cross-terms), the equation expands as follows:

$$\begin{aligned} \left\|\sum_{p\in\mathcal{S}_k} w_p \cdot \bar{\mathbf{v}}_p\right\|_2^2 &= \sum_{l=1}^{d_h}\left(\sum_{p\in\mathcal{S}_k} w_p \cdot \bar{v}_p^l\right)^2 \\ &\approx \sum_{l=1}^{d_h}\sum_{p\in\mathcal{S}_k}\left(w_p \cdot \bar{v}_p^l\right)^2 \\ &= \sum_{p\in\mathcal{S}_k} w_p^2 \left(\sum_{l=1}^{d_h}(\bar{v}_p^l)^2\right) \\ &= \sum_{p\in\mathcal{S}_k} w_p^2 \left\|\bar{\mathbf{v}}_p\right\|_2^2 . \end{aligned} \tag{16}$$

Consequently, the problem of identifying the optimal subset $\mathcal{S}_k$ that minimizes the attention output error reduces to maximizing the following objective, as presented in Section 3.2:

$$\boxed{\max_{\mathcal{S}_k} \sum_{p\in\mathcal{S}_k} \exp\left(\frac{A_p}{\sqrt{d_h}}\right)^2 \|\bar{\mathbf{v}}_p\|_2^2} \tag{17}$$

B Detailed Experimental Setup

B.1 Evaluation Benchmarks

We benchmark performance on two long-context evaluation suites, **RULER** (Hsieh et al., 2024) and **LongBench** (Bai et al., 2023). For RULER, we report the average score over 13 subtasks, each evaluated on 50 samples. For LongBench, we report the average score across all 16 English tasks.

B.2 Experimental Settings

Experiments are conducted across a wide range of sequence lengths and KV budgets. Specifically, we evaluate sequence lengths of 4K–16K for LongChat and Mistral, and 8K–32K for the Llama-3 and Qwen-2.5 families, while varying the token budget $C \in \{256, 512, 1024, 2048\}$.

For brevity, we use simplified model names in the main paper: LongChat-7B for *LongChat-v1.5-7B*, Mistral-7B for *Mistral-7B-v0.2-Instruct*, Llama-3.1-8B and Llama-3.2-3B for *Llama-3.1-8B/3.2-3B-Instruct*, and Qwen-2.5-7B and Qwen-2.5-3B for *Qwen-2.5-7B/3B-Instruct*.

B.3 Baselines and Implementation Details

We compare against **Full KV Cache** attention and **QUEST** (Tang et al., 2024), which performs query–key–only page selection. Our AOA-KV implementation extends the publicly available QUEST codebase. Following prior work, sparse attention is disabled for the first two layers, and the page size is fixed to 16. In addition, the first sink page and the last local page are always included within the KV budget to ensure stable attention behavior.

For models using GQA, allowing different heads within the same group to select different pages can reduce effective sparsity. To address this, we aggregate head-level criticality scores within each group and enforce a shared page selection, preserving both sparsity and efficiency.

B.4 Evaluation Metric

For Section 5.1, attention output error (AOE) is computed as the layer-averaged squared L_2 error

over the first 10 decoding steps. For Section 5.3, end-to-end decoding latency is averaged over five runs.